

Label Propagation Community Detection

Student Social Network — Neo4j GDS · GDS-45

1. Objective

This document presents the results of applying the Label Propagation Algorithm (LPA) to the student friendship network. LPA is a fast, near-linear community detection method that works by propagating labels through the graph: each node initially has a unique label, then iteratively adopts the most frequent label among its neighbors. Unlike modularity-based methods (Louvain/Leiden), LPA does not optimize a global objective; it tends to find communities that are locally cohesive. However, it is known to often converge to a single community in dense or highly connected networks, as observed here.

2. Implementation

Graph Projection

```
CALL gds.graph.project(  
  'student-friendship-graph-lpa',  
  'Student',  
  { EST_AMI_DE: { orientation: 'UNDIRECTED' } }  
)  
  
YIELD graphName, nodeCount, relationshipCount;
```

Label Propagation — Stream

```
CALL gds.labelPropagation.stream(  
  'student-friendship-graph-lpa'  
)  
  
YIELD nodeId, communityId;
```

Label Propagation — Stats

```
CALL gds.labelPropagation.stats(  
  'student-friendship-graph-lpa'  
)  
  
YIELD communityCount;
```

Label Propagation — Write

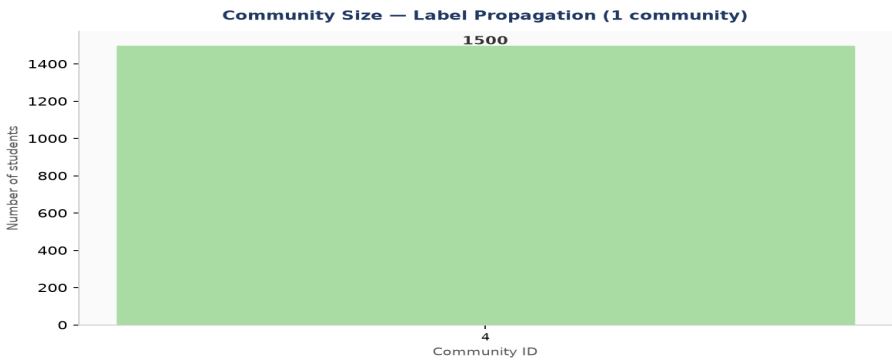
```
CALL gds.labelPropagation.write(  
  'student-friendship-graph-lpa',  
  { writeProperty: 'labelPropagationCommunity' }  
)  
  
YIELD communityCount, nodePropertiesWritten;
```

3. Global Results

Metric	Value
Number of communities	1
Nodes written	1,500
Community ID assigned	4 (internal node ID)

4. Community Size Analysis

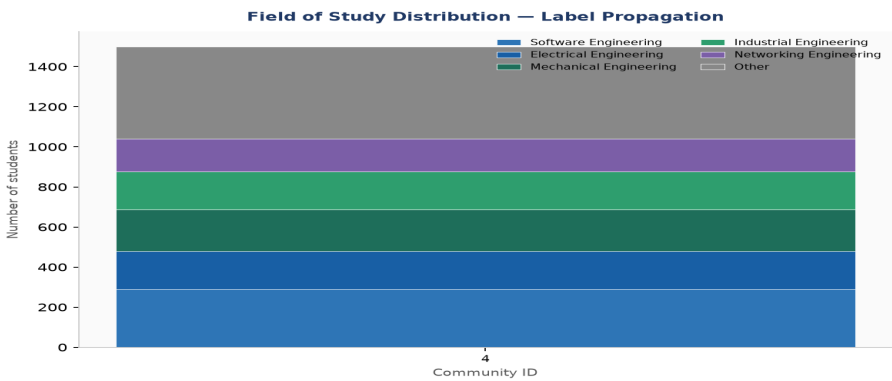
Label Propagation produced exactly one community containing all 1,500 students. The community ID (4) is not a sequential label but the internal Neo4j node ID of the representative node chosen by the algorithm (this is standard GDS behavior for LPA). The single community result indicates that, under the default parameters, the algorithm could not find a partitioning — all nodes converged to the same label.



Community ID	Size	% of total
4	1500	100.0%

5. Field of Study Distribution per Community

The stacked bar below shows the composition of the single community by field of study. The distribution mirrors the overall student population (30% 'Other', ~19% Software Engineering, etc.), confirming that the community is not biased toward any particular department — it simply encompasses everyone.



Field distribution details

Community	Software Eng.	Electrical Eng.	Mechanical Eng.	Industrial Eng.	Networking Eng.	Other	Total
4	290	190	208	189	162	461	1500

6. Observations & Analysis

- **Single community detected**

Label Propagation produced only one community containing all 1,500 students. This is a known behavior of LPA when the network is dense or when the algorithm converges to a consensus label. In such cases, the algorithm does not provide a useful partition of the network.

- **No modularity score**

LPA does not optimize modularity, so no modularity value is reported. This contrasts with Louvain and Leiden, which provide Q scores to judge partition quality.

- **Community ID is an internal node ID**

The assigned community ID (4) is the Neo4j internal ID of the node that became the dominant label. This is a technical artifact and does not carry semantic meaning.

- **Field composition matches global population**

The single community contains students from all fields in proportions very close to the overall dataset: Other (30.7%), Software Engineering (19.3%), Mechanical (13.9%), Electrical (12.7%), Industrial (12.6%), Networking (10.8%). This confirms that the community is not formed by academic department, but rather by the lack of any division.

- **LPA limitations**

LPA is sensitive to the initial labeling and can produce different results per run. It often fails to find meaningful subdivisions in well-connected graphs. For this network, the single-community outcome suggests that either the graph is too cohesive or that parameters (e.g., `maxIterations`, `weightProperty`) need tuning.

- **Recommendation system implication**

Since LPA did not partition the network, the ``labelPropagationCommunity`` property cannot be used as a meaningful feature for recommendations. However, this negative result is valuable: it indicates that the friendship network is extremely interconnected, and that other algorithms (Louvain/Leiden) are more appropriate for community detection.

7. Conclusion

The Label Propagation Algorithm applied to the 1,500-student friendship network resulted in a single community containing every student. This outcome, while not useful for segmentation, provides important insight into the network's structure: it is densely connected, causing LPA to converge to a global label. For community detection in this graph, modularity-based algorithms (Louvain, Leiden) are better suited and have already produced meaningful partitions (18 and 15 communities respectively). LPA will not be integrated into the recommendation system, but the analysis confirms the choice of the other algorithms for downstream tasks.